Ch 14: Probability

Question Bank | 2M | 3M | 5M

<u>SECTION A - 2 MARKS</u>

Q1. A die is thrown once. Find the probability of getting a prime number.

Q2. A die is thrown once. Find the probability of getting a number which is not a factor of 36. *(PYQ)*

Q3. A die is thrown once. Find the probability of getting a number between 2 and 6.

Q4. A die is thrown once. Find the probability of getting an odd number greater than 3.

Q5. Two dice are thrown simultaneously. Find the probability of getting a doublet.

Q6. Two dice are thrown simultaneously. Find the probability that the sum of numbers appearing on the two dice is more than 9.

Q7. Two dice are thrown at the same time. What is the probability that the sum of two numbers appearing on the top is more than 8? *(PYQ)*

Q8. Two dice are thrown simultaneously. Find the probability of getting a sum that is a multiple of 3.

Q9. Two dice are thrown simultaneously. Find the probability that the product of the numbers appearing on the top is less than 9.

Q10. A coin is tossed twice. Find the probability of getting at least one head. *(PYQ)*

Q11. Three coins are tossed simultaneously. Find the probability of getting exactly two heads.

Q12. Three coins are tossed simultaneously. Find the probability of getting at least two tails.

Q13. Three coins are tossed simultaneously. Find the probability of getting no head.

Q14. A card is drawn at random from a well-shuffled deck of 52 playing cards. Find the probability of getting a red card.

Q15. A card is drawn at random from a deck of 52 cards. Find the probability that it is a face card. *(PYQ)*

Q16. A card is drawn at random from a deck of 52 cards. Find the probability of getting a red face card. *(PYQ)*

Q17. A card is selected at random from a well-shuffled deck of 52 cards. Find the probability of getting a black king. *(PYQ)*

Q18. A card is drawn at random from a deck of 52 cards. Find the probability that it is neither a king nor a queen.

Q19. One card is drawn at random from a well-shuffled deck of 52 cards. Find the probability that the card drawn is not a spade.

Q20. From a well-shuffled deck of 52 cards, a card is drawn. Find the probability that the card drawn is a 10 of a red suit.

Q21. In a lottery, there are 10 prizes and 30 blanks. Find the probability of winning a prize. ***(PYQ)***

Q22. A bag contains 3 red, 5 white, and 7 blue balls. A ball is drawn at random. Find the probability that the ball drawn is blue.

Q23. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls.

Q24. A bag contains 6 red balls, 8 white balls, and 4 green balls. A ball is drawn at random. Find the probability that it is not green.

Q25. A jar contains 24 marbles — some are green and others blue. If a marble is drawn at random, the probability that it is green is 2/3. Find the number of blue marbles. ***(PYQ NCERT)***

Q26. If P(E) = 0.05, what is the probability of not E? ***(PYQ NCERT)***

Q27. An event E has probability 0.38. Find P(not E). Also explain what complementary events mean.

Q28. A box contains 20 balls numbered 1 to 20. A ball is drawn at random. Find the probability that the number on the ball is divisible by both 2 and 3.

Q29. A die is thrown once. Find the probability of getting a number which is a perfect square.

Q30. Cards numbered 1 to 30 are put in a box. A card is drawn at random. Find the probability that the number on the drawn card is a prime number less than 10.

Q31. The probability that it will rain on a particular day is 0.64. Find the probability that it will not rain on that day. Also find P(E) + P(not E) and verify.

Q32. A bag contains cards numbered from 1 to 25. A card is drawn at random. Find the probability that the number is a perfect square.

Q33. Two players A and B play a game. P(A wins) = 1/3. Find P(B wins) if the game cannot end in a draw.

Q34. A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought?

Q35. Write the sample space when two coins are tossed simultaneously.

Q36. What is the probability of an impossible event? Give one example.

Q37. What is the probability of a sure event? Give one example.

Q38. The probability of selecting a rotten apple from a heap of 900 apples is 0.18. Find the number of rotten apples in the heap.

SECTION B - 3 MARKS

Q1. Two dice are thrown simultaneously. Find the probability of getting: (i) a doublet (ii) a sum of 8 (iii) a sum divisible by 2 or 3. *(PYQ 2019)*

Q2. Two dice are thrown simultaneously. Find the probability of getting: (i) an even number on both dice (ii) a total of at least 10 (iii) the same number on both dice.

Q3. Two dice are thrown at the same time. Find the probability of getting: (i) a sum less than 7 (ii) a product less than 16 (iii) a doublet of odd numbers. *(PYQ)*

Q4. A pair of dice is thrown once. Find the probability of getting: (i) the same number on each die (ii) a sum greater than 9 (iii) a prime number on each die. *(PYQ 2020)*

Q5. Three coins are tossed simultaneously. Find the probability of getting: (i) exactly two heads (ii) at least one tail (iii) no head. *(PYQ)*

Q6. A card is drawn at random from a well-shuffled deck of 52 cards. Find the probability that it is: (i) a king of red colour (ii) a face card (iii) not a jack. *(PYQ 2019)*

Q7. From a pack of 52 cards, a card is drawn at random. Find the probability of getting: (i) an ace (ii) a red queen (iii) a black face card. ***(PYQ)***

Q8. From a pack of 52 cards, a card is drawn at random. Find the probability of getting: (i) neither a king nor a queen (ii) a spade or an ace (iii) a card that is not a heart. ***(PYQ 2020)***

Q9. Cards marked with numbers 3, 4, 5, ..., 50 are placed in a box and mixed thoroughly. A card is drawn at random. Find the probability that the number on the card is: (i) divisible by 7 (ii) a perfect square (iii) a prime number less than 20. ***(PYQ 2019)***

Q10. Cards marked with numbers 5 to 20 are placed in a box. A card is drawn at random. Find the probability that the number on the card is: (i) a prime number (ii) divisible by 3 (iii) an odd number. ***(PYQ)***

Q11. A box contains 20 cards numbered 1 to 20. A card is drawn at random. Find the probability that the number on the card is: (i) a multiple of 4 (ii) not divisible by 6 (iii) a number between 7 and 15. ***(PYQ)***

Q12. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. Find the probability that the ticket drawn has a number which is: (i) a multiple of 3 or 5 (ii) a multiple of 3 and 5. ***(PYQ)***

Q13. A bag contains 5 red balls, 8 white balls, and 4 green balls. One ball is drawn at random from the bag. Find the probability

that the ball drawn is: (i) white (ii) not green (iii) red or white. ***(PYQ 2018)***

Q14. A bag contains 15 white and some black balls. If the probability of drawing a black ball from the bag is thrice that of drawing a white ball, find the number of black balls. ***(PYQ)***

Q15. A bag contains 6 red, 8 white, and 4 green balls. A ball is drawn at random. Find the probability that it is: (i) red (ii) not green (iii) either white or green. ***(PYQ 2020)***

Q16. A jar contains 24 marbles. Some are green and others are blue. If a marble is drawn at random, P(green) = 2/3. Find: (i) the number of green marbles (ii) the number of blue marbles (iii) the probability of getting a blue marble. ***(PYQ NCERT)***

Q17. Red queens and black jacks are removed from a deck of 52 cards. A card is drawn at random. Find the probability of getting: (i) a king (ii) a red card (iii) a face card. ***(PYQ 2019)***

Q18. All aces, jacks, and black queens are removed from a pack of 52 cards. A card is drawn at random. Find the probability of getting: (i) a face card (ii) a red card (iii) a spade. ***(PYQ)***

Q19. Kings, queens, and jacks of clubs are removed from a deck of 52 cards. A card is drawn from the remaining cards. Find the probability of getting: (i) a heart (ii) a club (iii) a king. ***(PYQ)***

Q20. Cards bearing numbers 2 to 30 are placed in a bag. A card is drawn at random. Find the probability that the card bears: (i) a two-digit number (ii) a number which is a perfect square (iii) a prime number. ***(PYQ)***

Q21. An integer is chosen at random from the first 200 integers. Find the probability that it is: (i) divisible by 6 (ii) divisible by 8 (iii) divisible by 6 or 8. ***(PYQ)***

Q22. A number x is selected at random from the numbers 1, 2, 3, and 4. Another number y is selected at random from the remaining three numbers. Find P(x > y). ***(PYQ 2020)***

Q23. Two customers Shyam and Ekta are visiting a shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any one day. Find the probability that both will visit the shop: (i) on the same day (ii) on consecutive days (iii) on different days. ***(PYQ NCERT)***

Q24. A number is selected at random from 1 to 30. Find the probability that it is: (i) a prime number (ii) a composite number (iii) neither prime nor composite. ***(PYQ)***

Q25. In a class of 60 students, 25 like cricket, 10 like both cricket and tennis, and 30 like tennis. A student is selected at random. Find the probability that the student: (i) likes cricket only (ii) likes tennis only (iii) likes neither cricket nor tennis. ***(PYQ HOTS)***

Q26. A game consists of tossing a coin 3 times and noting the outcome each time. Hanif wins if he gets three heads or three tails. Find the probability that Hanif will lose the game. ***(PYQ NCERT)***

Q27. A die is numbered in such a way that its faces show the numbers 1, 2, 2, 3, 3, 6. It is thrown two times and the sum of the

numbers appearing is noted. Find the probability of getting each sum from 2 to 12. ***(PYQ NCERT)***

Q28. Suppose you drop a die at random on the rectangular region shown. What is the probability that it will land inside the circle with diameter 1 m if the rectangle is 3 m × 2 m?

Q29. All the face cards from a pack of 52 playing cards are removed. From the remaining pack, a card is drawn at random. Find the probability of getting: (i) a black card (ii) a card with number less than 5 (iii) a card numbered between 3 and 7. ***(PYQ)***

Q30. A coin is tossed three times. Find the probability of getting: (i) at most 2 heads (ii) exactly 1 tail (iii) at least 2 tails. ***(PYQ)***

Q31. A child has a die whose six faces show the letters given below. The die is thrown once. What is the probability of getting: (i) A (ii) D?***(PYQ NCERT)***

Face	**A**	**B**	**C**	**D**	**E**	**A**

Q32. Joel has 6 coins of Rs. 2, 5 coins of Rs. 10, and 3 coins of Rs. 20 in his pocket. He draws one coin randomly. Find the probability that: (i) the coin is sufficient to pay Rs. 7 (ii) the coin is of Rs. 20. ***(PYQ competency-based)***

Q33. The probability of selecting a green marble from a bag is 1/4 and the probability of selecting a blue marble is 1/3. Find the

probability of selecting neither a green nor a blue marble if the bag contains only green, blue, and red marbles.

Q34. The probability that a student passes in Mathematics is 2/3 and the probability that he passes in English is 4/9. If the probability of passing at least one subject is 4/5, find the probability that he passes in both subjects.

Q35. Savita and Hamida are friends. What is the probability that both will have: (i) different birthdays (ii) the same birthday — ignoring leap year? ***(PYQ NCERT)***

SECTION C - 5 MARKS

Q1. Two dice are thrown simultaneously. Complete the table below and find: (i) P(sum is even) (ii) P(sum is a prime) (iii) P(sum is a multiple of 3) (iv) P(sum is 1) (v) P(sum is less than or equal to 12). ***(PYQ 2019 — most repeated)***

Q2. A card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting: (i) a king of red colour (ii) a face card (iii) a red face card (iv) the jack of hearts (v) a spade (vi) the queen of diamonds. ***(PYQ NCERT — most repeated)***

Q3. All red face cards are removed from a pack of 52 playing cards. The remaining cards are well-shuffled and then a card is drawn at random. Find the probability that the drawn card is:

(i) a red card (ii) a face card (iii) a card of clubs (iv) not a king (v) an ace. ***(PYQ 2019)***

Q4. From a deck of 52 cards, the following are removed: all spades, face cards of heart, and all aces. A card is drawn at random from the remaining cards. Find the probability of getting: (i) a heart (ii) a black card (iii) a face card (iv) an ace (v) a red card. ***(PYQ HOTS)***

Q5. A bag contains cards numbered from 1 to 49. A card is drawn at random. Find the probability of getting: (i) a perfect square number (ii) a multiple of 5 (iii) an even prime number (iv) a number divisible by both 4 and 6 (v) a number less than 12. ***(PYQ 2020)***

Q6. Numbers 1 to 50 are written on separate slips (one number on each slip), kept in a box, and mixed well. One slip is drawn from the box. Find the probability of getting: (i) a number divisible by 5 (ii) a perfect cube (iii) a prime number between 10 and 30 (iv) a number not divisible by 3 (v) a two-digit number. ***(PYQ)***

Q7. A number is chosen at random from the numbers −3, −2, −1, 0, 1, 2, 3. Find the probability that the number is: (i) negative (ii) zero (iii) a non-negative number (iv) a number less than 3 (v) a number greater than −2. ***(PYQ)***

Q8. Two dice are thrown simultaneously. Find the probability of getting: (i) the sum as a prime number (ii) a total of at least 10 (iii) a doublet of even numbers (iv) a multiple of 2 on one die

and a multiple of 3 on the other (v) same number on both dice. ***(PYQ 2020)***

Q9. Case Study: A game at a fair involves spinning a wheel with equal sections numbered 1 to 8. Find the probability of: (i) getting an even number (ii) getting a number greater than 5 (iii) getting a multiple of 3 (iv) getting a prime number (v) getting the number 8. ***(PYQ 2023 case-based)***

Q10. A box contains 100 envelopes of which 20 contain Rs. 100, 35 contain Rs. 50, 30 contain Rs. 20, and 15 are empty. An envelope is drawn at random. Find the probability of getting an envelope that contains: (i) Rs. 100 (ii) Rs. 50 (iii) more than Rs. 20 (iv) money (v) nothing. ***(PYQ)***

Q11. There are 1000 sealed envelopes in a box, 10 of them contain a cash prize of Rs. 100 each, 100 of them contain a cash prize of Rs. 50 each, and 200 of them contain a cash prize of Rs. 10 each, and the rest do not contain any cash prize. If they are well shuffled and an envelope is picked up, find the probability of getting: (i) a cash prize (ii) a cash prize of Rs. 10 (iii) no cash prize (iv) a cash prize of Rs. 100 (v) a prize of more than Rs. 50. ***(PYQ)***

Q12. Cards marked with numbers 3 to 102 are placed in a box and mixed thoroughly. A card is drawn at random. Find the probability that the number on the card is: (i) a perfect square (ii) a multiple of 9 (iii) divisible by 10 and 6 both (iv) a prime number less than 25 (v) a two-digit number less than 30. ***(PYQ)***

Q13. A survey of 364 children aged 19–36 months was taken to determine the relationship between breast-feeding and respiratory illness. The data is tabulated below:

	Breast-fed	**Not breast-fed**
Respiratory illness	**42**	**90**
No respiratory illness	**150**	**82**

Find the probability that a randomly selected child: (i) had respiratory illness (ii) was not breast-fed (iii) was breast-fed and had no respiratory illness (iv) had no respiratory illness (v) was breast-fed. ***(PYQ HOTS)***

Q14. A number is selected randomly from 1 to 100. Find the probability that it is: (i) divisible by 10 (ii) a perfect square (iii) a prime number less than 50 (iv) divisible by 4 and 6 both (v) not divisible by 5. ***(PYQ)***

Q15. A bag contains 18 balls out of which x balls are red. (i) If one ball is drawn at random, what is the probability that it is a red ball? (ii) If 2 more red balls are put in the bag, find the new probability of getting a red ball. (iii) If the second probability is double the first, find x. (iv) Find the probability of getting a non-red ball after adding 2 more balls. (v) Verify P(E) + P(not E) = 1. ***(PYQ 2019 — most repeated)***

Q16. Case Study: Fifteen cards numbered 1 to 15 are placed in a bag. A card is drawn at random. Find the probability that the card drawn has: (i) a number less than 10 (ii) a prime number (iii) an odd number divisible by 3 (iv) a composite number less than 12 (v) a number which is neither prime nor composite. ***(PYQ 2024 case-based)***

Q17. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears: (i) a two-digit number (ii) a perfect square (iii) a number divisible by 5 (iv) a prime number less than 23 (v) a number divisible by both 5 and 3. ***(PYQ NCERT)***

Q18. A piggy bank contains 100 fifty-paise coins, 50 one-rupee coins, 20 two-rupee coins, and 10 five-rupee coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, find the probability that the coin which falls out: (i) will be a fifty-paise coin (ii) will be of value more than Rs. 1 (iii) will be of value less than Rs. 5 (iv) will be of value Rs. 1 or Rs. 2 (v) will not be a five-rupee coin. ***(PYQ NCERT)***

Q19. A game consists of tossing a one-rupee coin 3 times and noting the outcome each time. Ramesh wins Rs. 5 if he gets a head on the first toss and loses Rs. 1 otherwise. Deepak wins Rs. 3 if he gets 3 heads and loses Rs. 2 otherwise. Find: (i) the sample space (ii) P(Ramesh wins) (iii) P(Deepak wins) (iv) P(both win) (v) P(neither wins). ***(PYQ HOTS)***

Q20. Case Study: During a medical camp, blood groups of 400 people were recorded:

Blood Group	A	B	AB	O
Number	130	110	80	80

A person is selected at random. Find the probability that the person has blood group: (i) A (ii) B (iii) AB (iv) O (v) not AB.
(PYQ 2022 case-based)